Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

FIFTH SEMESTER B.TECH. (ELECTRICAL &ELECTRONICS)

END SEMESTER EXAMINATIONS, NOVEMBER-DECEMBER 2024

END SEMESTER EXAMINATIONS, NOVEMBER 2024

SUBJECT: COMMUNICATION SYSTEMS [ELE 3126]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: November 2024** **MAX. MARKS: 50**

Instructions to Candidates:

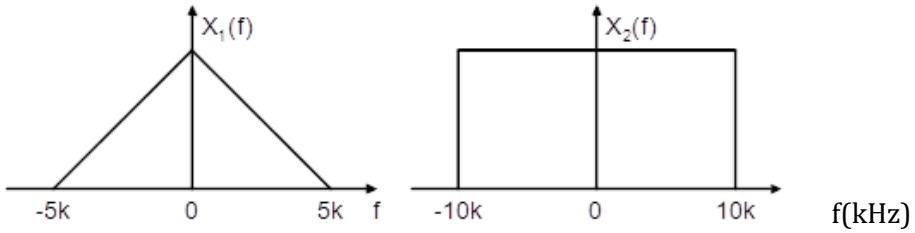
- Answer **ALL** questions.
- Missing data may be suitable assumed.

1A. Two signals $x_1(t)$ and $x_2(t)$ having fourier transform $X_1(f)$ and $X_2(f)$ shown in fig, are combined to form $y(t) = x_2(t) + 2x_1(t) \cos 2\pi f_c t$, fc = 20 kHz. If y(t) is transmitted then ***(04)***

a. Sketch the spectrum of the signal y(t).

b. Find the Bandwidth required for transmission

c. At the receiver side, suggest suitable method to recover $x_1(t)$ and $x_2(t)$ from y(t).



1B. The input to a square law modulator is V_i given by V_i=10 cos 2000πt + 0.05cos200πt. Assume the characteristics of the nonlinear device used to be $V_0 = V_i + 0.1V_i^2$. Then it is passed through a bandpass filter (BPF) with frequency range of 900Hz to 1100Hz. Compute the expression for the BPF output. ***(03)***

1C. In an Armstrong modulator, the crystal oscillator frequency is 200 KHz. The maximum modulation index in the phase modulator is 0.2 radians. The system is to accommodate modulation frequency 40Hz. At the output of the final multiplier the carrier frequency is to be 108MHz and the frequency deviation is 80KHz. Assume 10MHz mixer oscillator frequency . Compute the multiplier constants. ***(03)***

2A. Describe the FM slope detector demodulation technique. ***(03)***

2B. An analog message signal is limited in its excursions to the range from -20 to +20 volts. If there are 8 quantization level, and corresponding codes are located at -17.5(0), -12.5(1), -7.5(2), -2.5(3), 2.5 (4),7.5(5), 12.5 (6), 17.5 (7) then sketch the unipolar NRZ representation of the PCM output for the given sample values: ***(03)***

Sample values: -6.1 7.5

2C. Eight independent message signals are sampled and time multiplexed using PAM. Six of the message signals are having a bandwidth of 4KHz and other two have bandwidth of 12KHz. ***(04)***

i)Compare the transmission bandwidth requirements and overall sampling rate for Synchronous TDM and Asynchronous TDM transmission, for the mentioned signals.

ii)Develop asynchronous PAM-TDM commutator arrangement for this application.

3A. A direct sequence spread BPSK system uses three D flip flops and feedback taps [1,3]. ***(04)***

a. Construct Linear Feedback Shift Register in the given configuration

b. Determine the PN sequence with initial state 110

c. Show the waveform of the spread code in polar NRZ format

3B. Consider the transmitted signal for a QPSK scheme as, ***(03)***

$S_i(t)= \left\{\sqrt{\frac{2E}{T}}\cos(2\pi f_c t + (2i-1)\frac{\pi}{4})\right\}$, where E –symbol energy, T- symbol duration and f_c – carrier frequency

a) Compute the coordinates of message points and encoded inputs with i=1,2,3,4, and then draw the signal constellation diagram.

b) Draw the QPSK waveform for the message signal 01111000 with respect to the phase obtained in part b (consider bit rate = carrier frequency).

3C. The binary sequence 101101 is applied to a DPSK transmitter. ***(03)***

a) Draw the block diagram of the DPSK transmitter to transmit the sequence.

b) Sketch the resulting waveform at the transmitter output.[assume reference bit =1]

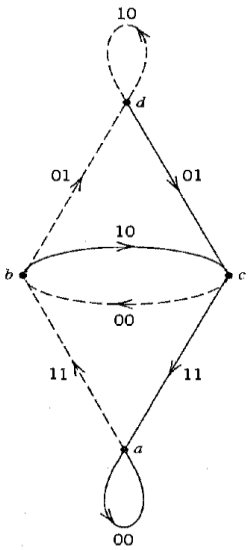
4A. A (6,3) linear block code is generated according to the generator matrix ***(03)***

$$G=\begin{bmatrix}1&0&1&1&0&0\\1&1&1&0&1&0\\1&1&0&0&0&1\end{bmatrix}$$

and the received code r=[1 1 0 1 1 1]. Decode this received word by finding the location of the error and obtain the correct code word.

4B. If the received sequence is [01101111], then using Viterbi algorithm, identify the message bits(decoded sequence), for a convolutional encoder with the given state diagram. ***(05)***

[where binary assignment of states are as given: a = 00, b = 10, c= 01 and d = 11]



4C. Let IEEE 802.11 standard uses slow FHSS for the Wi-Fi technology with the given specifications. If the input binary sequence transmitted is 01110010 and PN sequence generated is 0011. Sketch the frequency variation for one complete period of the PN sequence in slow FHSS. [Let number of MFSK tones (M)= 4 and length of PN segment per hop(k) = 2, Assume that the carrier is hopped to a new frequency after transmitting two symbols] ***(02)***

5A. Describe the symmetric cryptographic system with block diagram. What will be the ciphered text if the string "armedforce" is given as input to the code of Vigenere cipher with keyword as "hello"? ***(04)***

5B. For an artificial light system with curtain control, have 2 ambient light sensors, 2 temperature sensors, a motor to control up and down movement of curtain, 2 lamps and NodeMCU as a controller. Develop a star topology wireless network scheme of the mentioned scenario. (Consider the devices used as ZigBee devices, hence assume appropriate device roles in the ZigBee network while representing the topology). ***(03)***

5C. Let Ethernet follows a hexadecimal MAC address format. If the address is sent out on line in Ethernet with binary values: ***(03)***
01011010 00010001 01010101 00011000 10101010 00001111 then,

a) Find the type of the address?
b) Find the hexadecimal equivalent of the given Ethernet line address?
c)Describe the channel access technique of Ethernet.